

Comprehensive Research Overview: IIT Bombay Climate Studies Faculty – Current and Recent Research Activities

This unified list distills the **most specific, current** research topics pursued by Climate Studies faculty at IIT Bombay (2021–2025). Each bullet captures a deeply detailed aspect of their ongoing work:

- **Prof. Vishal Dixit** – quantifying latitude-dependent convective inhibition and advective–radiative coupling in Indian summer monsoon onset; kinetic-energy budget analysis of cross-equatorial Somali Jet formation and transition; subtropical shallow-convection momentum counter-gradient profiles in ICON-LEM; radiative-heating-driven ITCZ width control via cloud-top longwave anomalies; meridional-wind shear influence on northward ITCZ propagation; zonal free-tropospheric temperature-gradient modulation by cloud-base moistening; rapid-onset deep convective-scale energy transfers across monsoon troughs.
- **Prof. S. Ravichandran** – high-Reynolds-number turbulent particle-laden flow simulations in tropical convective systems using GPU-accelerated spectral solvers; coupled Lagrangian microphysical modeling of cloud droplet collision–coalescence in deep monsoon clouds; zonal flow instability pathways in tropical easterly jet regimes; scalable parallel implementations of two-phase Navier–Stokes for stratocumulus entrainment in LES; thermodynamic response functions of mixed-phase cirrus in radiative-convective equilibrium.
- **Prof. Angshuman Modak** – emergent-constraint derivation of equilibrium climate sensitivity accounting for dataset-dependent pattern-effect uncertainties; teleconnected aerosol-indirect forcing enhancement in polar amplification regions; machine-learning-augmented estimation of effective radiative-forcing response to quadrupled CO₂ in CMIP6; feedback parameter decomposition linking land–atmosphere radiative coupling to regional drought resilience; quantification of intermodel dependency in pattern-effect-sensitivity regressions; emergent constraint on aerosol-cloud-radiative interactions using multi-satellite retrieval ensembles.
- **Prof. Raghu Murtugudde** – high-resolution coupled ocean–atmosphere modeling of Bay of Bengal salinity stratification impacts on monsoon intraseasonal oscillations; predictive skill enhancement for extreme marine heat-wave occurrences via ensemble Kalman filter assimilation of SST fingerprint fields; parameterization of submesoscale nutrient upwelling in biogeochemical carbon-cycle simulations; Earth-system model downscaling for coastal flooding risk under compound cyclone–storm-surge events;

machine-learning correction of model structural biases in ENSO teleconnections affecting Indian precipitation.

- **Prof. Akshaya Nikumbh** – synoptic-scale blocking high dynamics modulating North Atlantic–Indian monsoon teleconnections during extreme central-India rainfall outbreaks; multiscale coupling between monsoon intra-seasonal oscillations and depressions producing localized versus widespread rainband events; spectral clustering of low-pressure synoptic modes driving super-monsoon break spells; spatial aggregation metrics for identifying shifts in sub-grid heavy-rain clusters; ground-based sky-imagery-derived cloud-fraction microphysical retrievals for validating satellite-based cumulus-microphysics feedbacks.